بسم الله الرحمن الرحيم

Chapter 2
Understanding Organizational Style and Its Impact on Information Systems

Systems Analysis and Design

Kendall and Kendall

Fifth Edition

Major Topics

- Organizational environment
- Nature of systems
- Context-level data flow diagram
- Entity-relationship diagram
- Levels of management

Organizations

- Organizations are composed of interrelated and interdependent subsystems
- System and subsystem boundaries and environments impact on information system analysis and design

Organizational Environment

- Community environment
  - Geographical
  - Demographics (education, income)
- Economic environment
  - Market factors
  - Competition
- Political environment
  - State and local government

Open and Closed Systems

- Systems are described as either
  - Open
    - Free-flowing information
    - Output from one system becomes input to another
  - Closed with restricted access to information
    - Limited by numerous rules
    - Information on a need to know basis

Virtual Organizations

- A virtual organization has parts of the organization in different physical locations
- Computer networks and communications technology are used to work on projects

Virtual Organization Advantages

- Advantages of a virtual organization are
  - Reduced costs of physical facilities
  - More rapid response to customer needs
  - Flexibility for employees to care for children or aging parents

Depicting System Graphically

Context-Level Data Flow Diagram (DFD)

- A context-level data flow diagram is an important tool for showing data used and information produced by a system
- It provides an overview of the setting or environment the system exists within: which entities supply and receive data/ information

Context-Level DFD Symbols

- Entity, a person, group, department, or system that supplies or receives information
- It is labeled with a noun

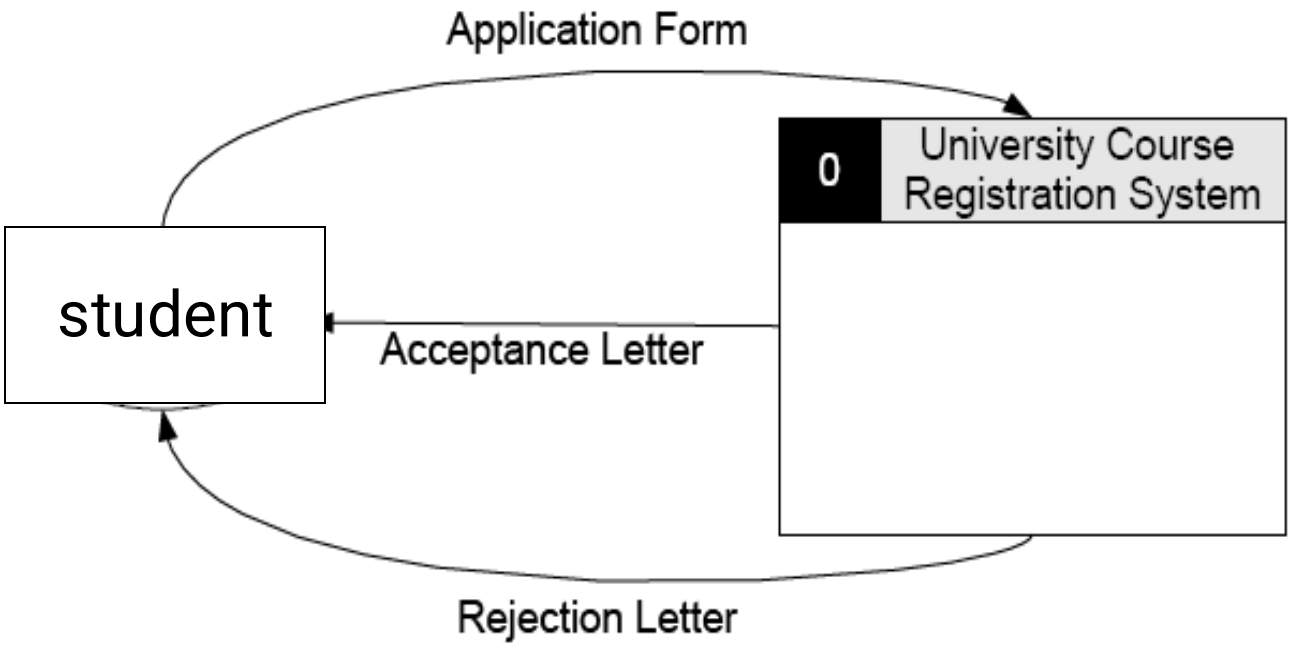
Context-Level DFD Symbols

- Process, representing the entire system
- It is given the number 0

Context-Level DFD Symbols

- Data flow, represented by an arrow
- It shows information that passes to or from the process
- Data flow is labeled with a noun

- **Context Diagram**



Entity-Relationship Diagrams (E-R Diagrams)

- Entity-relationship diagrams help the analyst understand the organizational system and the data stored by the organization
- Symbols are used to represent entities and relationships

Fundamental Entity

- Describes a person, place, or thing
- Symbol is a rectangle

Relationships

- Relationships show how the entities are connected
- There are three types of relationships:
  - One to one
  - One to many
  - Many to many
- Relationship lines are labeled

Relationship Notation

- One is indicated by a short vertical line
- Many is indicated by a crows foot

Attributes

- Data attributes may be added to the diagram

Creating Entity-Relationship Diagrams

- Steps used to create E-R diagrams
  - List the entities in the organization
  - Choose key entities to narrow the scope of the problem
  - Identify what the primary entity should be
  - Confirm the results of steps 1-3 through data gathering methods (interviewing, questionnaire, observation)

Levels Of Management

- The three levels of managerial control are
  - Operations management
    - Make decisions using predetermined rules
  - Middle management
  - Strategic management

Operations Management

- Make decisions using predetermined rules that have predictable outcomes
- Work is clear-cut
- High degree of certainty in decision making
- Oversee the operating details of the organization
- Dependent on internal information

Middle Management

- Make short-term planning and control decisions about resources and organizational objectives
- Experience very little certainty in their decision making
- Partly operational and partly strategic
- Dependent on internal information, both historical and prediction oriented

Strategic Management

- Look outward from the organization to the future
- Make decisions that will guide middle and operations managers
- Work in highly uncertain decision-making environment
- Define the organization as a whole
- Dependent on external information

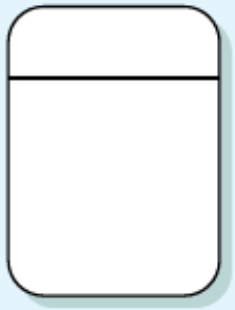
Managerial Levels

- Each of the three levels of management have
  - Different organization structure
  - Leadership style
  - Technological considerations
  - Organization culture
  - Human interaction
  - All carry implications for the analysis and design of information systems

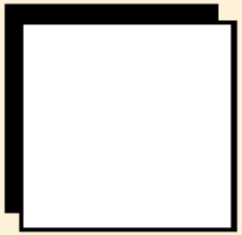
Context-Level Data Flow Diagrams

- Focus is on the data flowing into and out of the system and the processing of the data
- Shows the scope of the system:
  - What is to be included in the system
  - The external entities are outside the scope of the system

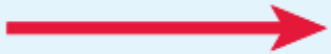
The Basic Symbols of a Data Flow Diagram (Figure 2.4)



A **process** means that some action or group of actions take place.



An **entity** is a person, group, department, or any system that either receives or originates information or data.



A **data flow** shows that information is being passed from or to a process.

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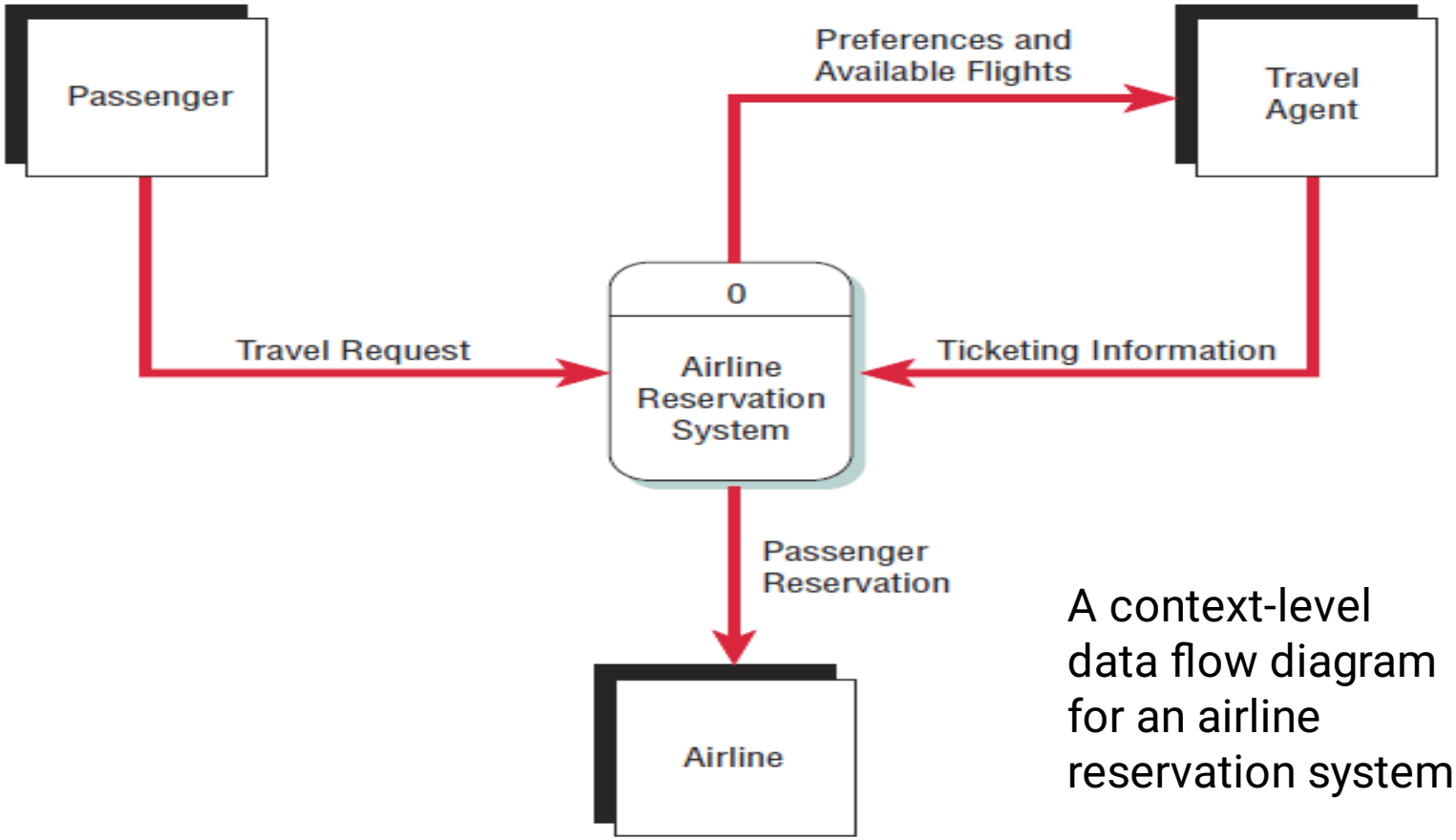
Airline Reservation System (Figure 2.5)



A context-level data flow diagram for an airline reservation system

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example

A student sends in an application form containing his personal details, and his desired course. The university checks with the course list whether the course is available. If the course is available the student is enrolled in the course, and the university confirms the enrollment by sending a confirmation letter that they are registered for the course to the student. Or if the course is unavailable the student is sent a rejection letter.

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